

ANSWERS

1.

Question	Answer	Marks
5(a)(i)	faster-moving <u>particles</u> and (more likely to) escape (from the water)	B1
	remaining <u>particles</u> have less (average) kinetic energy / move more slowly (on average)	B1
5(a)(ii)	(work is done) moving particles apart (against the forces of attraction) or (energy needed) to supply the latent heat	B1
5(b)	any two from: no bubbles formed or occurs only at the surface occurs at any temperature is affected by exposed surface area / moving air over surface / temperature	B2

2.

Question	Answer	Marks
4(a)(i)	–273 (°C) and 0 (K)	B1
4(a)(ii)	particles) move fast(er)	B1
	<u>vibrate</u> / <u>oscillate</u> with larger amplitude	B1
4(a)(iii)	internal energy increases	B1
	temperature stays the same	B1
4(b)(i)	($E = mcT$) in any form numerical or algebraic	C1
	8000 (J)	A1
4(b)(ii)	100–44 or 56	C1
	0.48 or 0.47 (J/(g °C))	A1

3.

Question	Answer	Marks
5(a)	ANY 2 from - as temperature rises / heat added particles move faster / vibrate faster / move further apart - smaller force between particles (in liquid)	B1
	regular / fixed arrangement of particles in a solid or more space between particles in a liquid / less tightly packed / particles further apart	B1
5(b)(i)	373 (K)	B1
5(b)(ii)	(energy needed) to <u>separate</u> particles / increase potential energy of particles / overcome (intermolecular) forces (of attraction)	B1
5(c)	$E = Pt$ in any form, algebraic or numerical, seen, e.g. 2400 (J)	B1
	(shc =) E / mT in any form	B1
	0.48 (J/(g °C))	B1

4.

Question	Answer	Marks
9(a)(i)	chemical	B1
9(a)(ii)	electrical	B1
9(a)(iii)	ANY 2 from - light - sound - radio / microwave / electromagnetic (waves)	B2
9(b)	($Q =$) It in any form	C1
	0.16 C	A1
9(c)(i)	($E =$) mcT numerical or algebraic in any form	C1
	$0.11 \times 830 \times 5$	C1
	460 J	A1
9(c)(ii)	energy output / energy input	C1
	useful energy output / (total) energy input	A1
9(c)(iii)	(energy input =) $460 + 5200$ or 5660 (J) seen or $5200 \div (\text{energy in } 9(c)(i) + 5200)$	C1
	92% or 0.92	A1
9(c)(iv)	(thermal) energy <u>lost</u> / heat <u>lost</u> (from battery to environment)	B1
	energy input larger than calculated / 5660 (J) or temperature rise should be larger / is smaller than it should be or thermal energy / heat produced larger than 460 J / calculated	B1

5.

Question	Answer	Marks
3(a)	($E =$) mcT in any form algebraic or numerical	C1
	$2 \times 460 \times 450$	C1
	4.1×10^5 J	A1
3(b)(i)	any one of: - boiling occurs at one temperature / boiling point or evaporation occurs at any temperature / below boiling point - boiling occurs within liquid or evaporation occurs on the surface - evaporation increased by draughts, more surface area (and boiling does not)	B1
3(b)(ii)	(average) speed / K.E. of molecules decreases (as temperature falls) or (molecules) do not have the energy to escape / break bonds	B1
	fewer fast / energetic molecules escape or (only) fastest / high KE molecules escape	B1

6.

Question	answer	Marks
9(a)	(oil) hotter initially or cooler finally	B1
	temperature difference (between oil and freezer) decreases	B1
	loses energy faster when hotter or more slowly when cooler	B1
9(b)	molecules move more slowly or less <u>kinetic</u> energy or less internal energy	B1
	level decreases and oil contracts / volume of oil decreases	B1
	oil / liquid contracts more than glass / solid or molecules move closer	B1
9(c)(i)	-10°C	B1
9(c)(ii)	(attractive intermolecular) force between molecules or liquid solidifies	B1
	(intermolecular) potential energy decreases (as molecules move closer)	B1
	thermal energy lost (to freezer) is potential energy lost or latent heat lost	B1
9(d)(i)	$(Q =) ml_f$ or $0.045 \times 5.7 \times 10^4$	C1
	$2.6 \times 10^3 \text{ J}$	A1
9(d)(ii)	0.36 J / s or 0.36 W	B1
9(e)	specific heat capacity of liquid oil is smaller (than of oil in solid state)	B1
	the temperature decreases more quickly (in liquid state)	B1

7.

Question	Answer	Marks
4(a)	(molecules in) <u>fixed</u> places / positions / arrangement / pattern	B1
	(molecules in) lattice or in an orderly / repeating / regular arrangement	B1
	(molecules) close together or tightly packed	B1
4(b)(i)	$(Q =) mc\Delta T$ or $8.0 \times 4200 \times (43 - 18)$ or $8.0 \times 4200 \times 25$	C1
	$8.4 \times 10^5 \text{ J}$ or $840\,000 \text{ J}$	A1
4(b)(ii)	$\Delta\theta = Q / C$ or $8.4 \times 10^5 / 850$ or $988 (^{\circ}\text{C})$ or $990 (^{\circ}\text{C})$ or $1013 (^{\circ}\text{C})$	C1
	$988 (^{\circ}\text{C})$ or $990 (^{\circ}\text{C})$ or $1013 (^{\circ}\text{C})$ or $43 + 8.4 \times 10^5 / 850$	C1
	$1031 ^{\circ}\text{C}$ or $1030 ^{\circ}\text{C}$ or $1033 ^{\circ}\text{C}$ or $1000 ^{\circ}\text{C}$	A1

8.

Question	Answer	Marks
3(a)	any three from: molecules / they are further apart forces between molecules / them smaller molecules / they move in straight lines (between collisions) potential energy of molecules / them greater gas molecules move freely/randomly but liquid molecules slide over each other	B3
3(b)	(molecules separated against) intermolecular forces / bonds	B1
	work is done as molecules separate or bonds broken or latent heat supplied	B1

9.

Question	Answer	Marks
4(a)	any two from: only occurs at the surface / no bubbles produced occurs at any temperature produces cooling	B2
4(b)	molecules escape (from liquid)	B1
	fast(est) molecules / molecules with great(est kinetic) energy escape or slow(est) molecules / molecules with small(est kinetic) energy remain or (average) speed of remaining molecules decreases or molecules gain (kinetic) energy and escape	B1
4(c)(i)	rate of evaporation increases (with increasing area)	B1
4(c)(ii)	rate of evaporation decreases with decreasing temperature or takes too much time	B1

10.

Question	Answer	Marks
10(a)(i)	<i>Pair 1</i> boiling occurs at one temperature / constant temperature evaporation occurs at all / any temperature	B1
	<i>Pair 2</i> boiling occurs within / throughout liquid or causes bubbles evaporation occurs at surface	B1
	<i>Pair 3</i> boiling requires a heat <u>source</u> / affected by rate of heating evaporation causes cooling / affected by wind etc.	B1
	<i>Pair 4</i> temperature remains constant during boiling evaporation temperature drops / liquid cools	B1
10(a)(ii)	(thermal energy) used to break bonds or work done against intermolecular forces	B1
	molecules move further apart or P.E. increases or (molecules) push back the atmosphere	B1

11.

Question	Answer	Marks
3(a)	any two of: irregular arrangement (of molecules) intermolecular forces weak(er) / not held as firmly together intermolecular distances greater / more spaced out move in clusters through the liquid (not just vibrations) or positions not fixed or can slide past each other	B2
3(b)(i)	work done or forces overcome	B1
	atoms pulled apart or bonds broken	B1
3(b)(ii)	(Q =)ml or 0.84×64 or $0.84 \times 64\,000$	C1
	54 kJ or 54 000 J	A1

12.

Question	Answer	Marks
10(a)(i)	$(I =) P / V$ or 2.8 / 230 or 2800 / 230	C1
	0.012 or 12	C1
	12 A	A1
10(a)(ii)	12 A < integral number of amperes < 20 A	B1
10(b)(i)	$(Q =) m\Delta Tc$ or $6.3 \times (49 - 23) \times 4200$ or $6.3 \times 26 \times 4200$	C1
	6.9×10^5 J	A1
10(b)(ii)	<u>more thermal</u> energy is lost (per unit time)	B1
10(c)(i)	(when heated) water expands / volume of water increases / water becomes less dense	B1
	heated water / water with more energetic particles rises	B1
	cooler / denser water sinks or circulation or convection current	B1
10(c)(ii)	two separate approaches I / II	
	I atoms / molecules / ions / particles vibrate (more violently)	B1
	collide with neighbouring atoms / molecules / ions / particles	B1
	energy / heat / vibration passed on (to neighbour)	B1
	or	
	II particles / atoms / molecules / ions vibrate (more violently)	B1
	atoms / molecules / ions / particles strike electrons	B1
10(c)(iii)	electrons travel through transporting energy / heat	B1
	<u>particles / molecules / atoms</u> move apart (on average)	B1
	volume increases and density decreases	B1

13.

Question	Answer	Marks
4(a)	$(c =) E / mT$ or 17000 / (22 × 850)	C1
	0.91 J / (g °C)	A1
4(b)	765 – 774 J / °C	B1
4(c)	molecules colliding against molecules or movement / diffusion / collision of (free) electrons	B1

- 14.
- (a) (i) molecules / they close together **or** small gaps between molecules B1
- (ii) molecules / they exert large (repulsive) forces B1
- (b) (i) $(V =)m/\rho$ **or** $680/0.85$ C1
 800 cm^3 **or** $8.0 \times 10^{-4}\text{ m}^3$ A1
- (ii) 1 molecules vibrate molecules vibrate B1
collide with neighbours **or** collide with electrons B1
transfer vibration / energy electrons travel through metal B1
- 2 heated / hot oil expands / less dense B1
rises B1
convection current / circulation established B1
- 3 any suitable named insulator **and** it is a poor conductor B1
- (c) (i) temperature at which (liquid) vaporises / becomes gas / steam B1
- (ii) $(Q =)mc \Delta T$ **or** $680 \times 2.0 \times (260 - 20)$ **or** $680 \times 2.0 \times 240$ C1
 $3.3 \times 10^5\text{ J}$ A1
- (iii) heat supplied to pan **or** heat lost to air / surroundings B1
- 15.
- (a) **temperature** at which liquid / water turns to gas / vapour / steam B1
- (b) (i) $(T =)24\text{ }(^{\circ}\text{C})$ **or** $100 - 24$ **or** 76 C1
 $(\Delta Q =)mc\Delta T$ **or** $1.5 \times 4200 \times 76$ C1
 $4.8 \times 10^5\text{ J}$ A1
- (ii) heat is lost (to the surroundings) **or** evaporation B1
at higher temperatures heat is lost at greater rate B1
- (c) (i) stays at $100\text{ }^{\circ}\text{C}$ / constant B1
- (ii) molecules separate / are pulled apart / are far apart / break bonds /
overcome forces of attraction B1
work done separating the molecules **or** molecules gain PE B1

- 16.
- (a) (i) temperature B1
when solid turns to liquid B1
- (ii) molecules escape (surface) C1
fastest molecules / most energetic molecules A1
escape / break bonds
leaving behind slower molecules / colder molecules B1
or temperature falls
- (b) (i) at the surface / top of liquid B1
- (ii) less heat / energy enters (liquid nitrogen) / transfers B1
or less nitrogen evaporates / boils B1
reduces / stops conduction **and** convection B1
explanation of no conduction or convection, e.g. no molecules / no medium
- 17.
- (a) **temperature** at which a liquid becomes a gas B1
- (b) (i) molecules close together / touching **or** closer than in gas B1
randomly arranged **or** irregular structure B1
- (ii) to separate / increase the distance between molecules B1
work done against (intermolecular) forces **or** supply p.e. **or** break bonds B1
- 18.
- (a) $(H=) mcT$ **or** $330 \times 4.2 \times 13$ C1
18 000 J **or** 18 020 J **or** 18 018 J A1
- (b) ice takes in / needs heat / energy B1
- for latent heat
 - to melt / turn to water (at 0 °C) / change state
 - to break bonds / for molecules to gain P.E.
- water (in jug initially at 0 °C) warms up B1
or ice (and melted water in jug)
- stays at 0 °C / stays cold / stays at constant temp.
 - gives larger temperature difference (between liquid and melting ice in jug)
- (c) metal is a good conductor (of heat) B1
or metal / can has lower heat capacity
allow opposite statements for plastic, e.g. plastic is an insulator (of heat),
penalise wrong statements and Physics, e.g. liquid evaporates from can, metals
conduct temperature / convect better

- 19.
- (a) (liquid) molecules not arranged (so) regularly B1
 (liquid) molecules not vibrating / moving in same direction B1
or do not have same speed
- (b) (i) molecules / liquid escape (from surface) / break bonds B1
- (ii) fast moving / more energetic molecules evaporate / escape B1
 leaving slow molecules **or** molecules with less **kinetic** energy (on average) B1
- (c) (i) hot air rises B1
- (ii) (steam) condenses or changes to liquid (on thermometer) B1
or heat (conducted) from hot to cold
 gives out latent heat (to thermometer) B1
or explanation involving bonds being made
- 20.
- (a) (the molecules) move faster **or** have more **kinetic** energy **or** accelerate **ignore** vibrate faster B1
- (b) (i) faster / energetic molecules escape B1
 average speed decreases **or** slower molecules remain B1
- temperature depends on average KE **or** heat taken from runner B1
OR
 liquid becomes gas / vapour
 latent heat needed or bonds broken
 heat taken from runner
- (ii) water vapour blown away **or** surrounding air less humid B1
- 21.
- (a) (i) any sensible example where expansion is useful [B1]
- (ii) any sensible example where expansion causes a problem [B1]
- (b) (molecules) move fast(er) or vibrate fast(er) or have more (kinetic / potential / internal) energy [B1]
- (molecules) move apart or distance between molecules increases or vibration has larger amplitude or vibration takes up more space or bonds stretch [B1]
- (c) slightly smaller [B1]
 much larger [B1]

- 22.
- (a) current **and** voltage /p.d./e.m.f. in correct order [B1]
- (b) (c=) E/mT in symbols or numbers e.g. $17\,000 = 0.85 \times c \times 22$ [C1]
 $910\text{ J/(kg }^\circ\text{C)}$ [A1]
- (c) (i) (hot air) rises **or** convection mentioned [B1]
 (hot) air less dense [B1]
- (ii) lag **or** cover with insulating material **or** warmer room **or** start with colder block [B1]
- 23.
- (a) ($m =$) ρV **or** 1000×450 C1
 $4.5 \times 10^5\text{ kg}$ A1
- (b) (i) ($Q =$) $mc\Delta T$ **or** $4.5 \times 10^5 \times 4.2 \times 15$ **or** 4200 **and** $15/(27-12)$ C1
 $4.5 \times 10^5 \times 4200 \times 15$ **or** $2.8(35) \times 10^7$ C1
 $2.8(35) \times 10^{10}\text{ J}$ A1
- (ii) thermal/internal energy/heat lost **or** gained by something specific (e.g. air/pool walls/tiles etc.) **or** heat lost by evaporation B1
- 24.
- (a) power supply with ammeter and heater in series B1
 voltmeter in parallel with heater/ power supply B1
- (b) (i) ($P =$) VI in any form C1
or 4.2×12
- $50(.4)\text{ W}$ A1
- (ii) ($E =$) Pt i.e. any power $\times$ any time e.g. $50(.4) \times 8$ C1
- $8/60$ C1
or 0.13(3) seen
or division by 1000 seen anywhere
- $0.0067(2)\text{ (kWh)}$ A1

(c) (i) molecules escape (from surface/leave water) / become gas or vapour / break bonds C1

fast(er) moving / high energy/ energetic molecules escape A1

(ii)

change	M1	explanation
wind / draught / breeze		wind knocks molecules away
or larger surface area		more chance/possibility of escape/more space to escape or more molecules come to/near/at surface
or decrease humidity / drier air		fewer molecules return/from air
or decrease atmospheric pressure		fewer air molecules to hit during escape

(iii) evaporation occurs at surface **and** boiling inside liquid/bubbles
 evaporation occurs at any temperature (accept room temperature)
and boiling occurs at boiling point/100°C/ fixed / specific temperature
 evaporation increased by draughts/higher temp/more area **and** boiling is not
OR increase in pressure stops boiling but only reduces evaporation
 any two B2

(d) water heats air (by conduction) B1
 or water loses heat/energy (to cup or air)
 or air gains heat/energy (from water)

hot / heated air / particles rise B1
 or cold air / particles sink
 or hot air is less dense
 or cold air is more dense

- 25.
- (a) 80 °C B1
- (b) $(Q =) mcT$ in any form numerical or algebraic C1
1530 J A1
- (c) (i) Any 2 lines from
- **latent** heat/energy mentioned
 - **latent** heat/energy given out/lost
 - bonds being made/strengthened
 - molecules lose PE
 - molecules KE constant
- B2
- (ii) ANY 2 lines but max 1 if no change/comparison implied
molecules change **from** OR in liquid
- random arrangement
 - move throughout in some form (e.g. move freely)
 - move or occur in clusters
- change **to** OR in solid
- regular arrangement/shape or fixed position/shape
 - vibrate
 - separation (probably) close(r)
- B2
- 26.
- (a) **temperature** when solid melts;
temperature when solid changes to liquid B1
- (b) (i) temperature increases; molecules move faster/have more kinetic energy/
vibrate faster/move further apart B1
- (ii) change of state; solid changes to liquid B1
latent heat provided; break bonds; molecules move apart/break free;
reduce bond strength; idea of more disorder B1
- (c) liquids expand more than solids B1

27.		
(a)	(i) molecules have more kinetic energy/speed/velocity hit sides hard(er)/with more force // (initially) hit sides (more) often/frequently // create large(r) pressure (initially)	B1 B1
	(ii) (larger) forces between liquid molecules/(stronger) bonds	B1
(b)	(i) $P_1 V_1 = P_2 V_2$ numerical or algebraic 6(.0) cm ³	C1 A1
	(ii) temperature is constant // no gas enters/leaves // mass constant	B1
28.		
(a)	(i) 50°C and 24/25°C	B1
	(ii) heat loss or evaporation mentioned // molecules escape // liquid to vapour more heat loss or more evaporation etc. because temperature is higher	C1 A1
	(iii) temperature becomes 100°C // reaches boiling point // temperature becomes steady water boils // water turns to steam/gas // energy loss = energy gain	B1 B1
(b)	(i) ($c =$) $E/m\Delta T$ in any form, numerical or algebraic, e.g. $7400/72 \times 23$ 4.5 or 4.47 or 4.4686 J/(g °C) // 4468.6 J/(kg °C)	C1 A1
	(ii) ($E =$) $\frac{1}{2} mv^2$ algebraic only $\frac{1}{2} \cdot 0.072 \cdot 450^2$ 7300 J // 7290 J (7 290 000 (J) alone gets 2/3)	C1 C1 A1
	(iii) <i>water molecules</i> move/vibrate fast(er)/(more) vigorously <i>water molecules</i> random motion // move (more) throughout liquid/all directions // slide over each other // move in convection // hit more often // move further apart <i>bullet molecules</i> motion in one direction/away from gun/towards target/ all have same (increase in) speed	B1 B1 B1
29.		
(a)	increased/high(er) temperature/hot(ter) wind or air flow less humidity less pressure	ANY 2 lines B2
	(b) molecules/atoms/particles escape/leave (surface) or molecules become gaseous/vapour or molecules break bonds molecules with large(est) energy/high(est) speed sufficient or enough energy escape/break bonds/become gaseous or leave behind slow/less energetic molecules	C1 A1

30.
 (a) mgh or $F \times d$ or 10×700 C1
 (-)7000 J A1
- (b) $Q/E/H = mc\Delta T$ or $(\Delta T =) 7000/(1) \times 4200$ C1
 1.7 or 1.67 or 5.5 C1
 8.9 °C e.c.f. (a) A1
31.
 (a) (i) atoms vibrate/move back and forth/to and fro (accept particles/molecules) M1
 atoms hit neighbours or pass on heat/energy to neighbour (not vibrations) A1
- (ii) atoms take up more space/further apart/larger vibrations (not atoms larger) B1
- (b) atoms move throughout (liquid) or not in fixed places or arrangement irregular or B1
 broken bonds (e.g. atoms move faster) B1
 atoms move at random/further apart (e.g. fixed volume/variable container shape etc.) B1
32.
 (a) rise in temperature / hot / heated B1
 road / bridge / rail / metal expands or gap reduces B1
 no buckling / deformation / breaking / cracking / twisting / tilting B1
- (b) any other problem + solution
 e.g. concrete cracks – leave a gap, telephone wires sag – put them high / tight
 hot water cracks glass – use thin glass / car engines seize up – cool them
 water freezes in pipes – lag them or use antifreeze / tyres burst – let air out
 pipes bend – use flexible joints / dashboard deforms – car in shade
 wrong readings on measuring cylinder – use correct temp. B1